Issy-les-Moulineaux, France
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JOEL SOFFO

Experience

Machine Learning Scientist (Ph.D CIFRE Program) — Issy-les-Moulineaux, France
AIRBUS — July 2023 - July 2026

- Research on data-driven methods for solving physics-based problems.
- Design and implementation of deep learning architectures for operator learning and Scientific ML.
- Developed dimensionality reduction techniques to accelerate high-dimensional simulation workflows.

Machine Learning Scientist — Issy-les-Moulineaux, France
AIRBUS — Sept. 2022 - July 2023

- Conducted applied research in Scientific Machine Learning.
- Implemented and benchmarked Neural Operators and PINNs for industrial simulation tasks.

Research Intern — Issy-les-Moulineaux, France
AIRBUS — Apr. 2022 - Sept. 2022

- Evaluated state-of-the-art methods in Scientific Machine Learning (PINNs, Neural Operators).
- Built experimental pipelines in PyTorch for PDE-based simulations.

Data Scientist Intern — Nemours, France
LABELIANS — Apr. 2021 - Sept. 2021

- Optimized pricing models through statistical analysis.
- Developed a user-facing analytics interface for marketing teams.

Education

Ph.D in Applied Mathematics — Nantes, France
Ecole Centrale de Nantes — 2023 - 2026

- Defense scheduled for September 2026.
- Advisor: Prof. Anthony Nouy
- Research interests: Deep Learning, dimensionality reduction methods, Physics-Informed Machine Learning, Operator Learning.

Master Data Science — Palaiseau, France
Institut Polytechnique de Paris (IP Paris) — 2021 - 2022

Publications

1. A. Bensalah, A. Nouy, **J. Soffo**. Nonlinear manifold approximation using compositional polynomial networks. *SIAM Journal on Scientific Computing*, 2026
2. M. Dus, G. Dusson, V. Ehrlacher, C. Guillot, **J. Soffo**. Comparison between tensor methods and neural networks for the computation of the Schrödinger equation ground state in electronic structure calculations. *ESAIM: Proceedings and Surveys*, 2024

Selected Projects

CEMRACS 2023

- Investigated deep learning methods to solve Schrödinger equation

Generative Modeling for Financial Risk 2022

- AI competition organized by BNP Paribas and Ecole Polytechnique
- Developed stress-testing models using GANs and VAEs

Ransomware Detection via Bitcoin Transactions 2022

- Built feature extraction pipeline from blockchain data
- Trained classification models to detect ransomware attack families

Awards and Honors

- **Third Prize**, Generative Modeling Data Science Challenge by BNP Paribas / X 2022
- **Best intern of the year**, LABELIANS 2021
- **Academic Scholarship**, Fondation Mathématique Jacques Hadamard (FMJH) 2020

Skills

Languages: French (native), English (fluent)

Programming: `Python`, `R`, `MATLAB`

Tools: `Git`, `VS Code`, `PyCharm`

Frameworks: `PyTorch`, `TensorFlow`, `Keras`, `scikit-learn`, `JAX`

Teaching

- Teaching Assistant: Mathématiques pour les Sciences - First year undergraduates, *Sorbonne Université (Paris-VI)* 2023-2024
- Teaching Assistant: Numerical and Fourier Series - Second year undergraduates, *ESIEE Paris* 2023-2024

Management

Internship supervisor of Takumi HADA (ENSTA 2024) - Internship subject: Machine Learning for simulation workflows